

Music-Gen v4 closure campaign — cycles 1-3

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Music-Gen v4 closure campaign — cycles 1-3

Abstract

Cycles 1-3 open the v4 closure campaign with the Chicken Grease bass sound-matching sub-milestone (M-V4-PROFILES-1/cg-bass), the first per-instrument search of the five-song profile-pinning milestone. The determinism certificate that gates palette-primary claims (M-V4-CERT) was already complete on disk when the run began: two `--no-cache` runs of the checkpointed driver produced byte-identical delivery WAVs and the certificate carries verdict `E2E_DETERMINISM_HOLDS` under `env_pin_sha256 = 623df01f...6571d38d`; that milestone landed trivially at cycle-1 opening.

The cg-bass arc then proceeded in three stages. Cycle 1 ran a coarse General-MIDI SoundFont sweep of 15 presets $\times$ 1 default configuration against the reference bass stem; program 17 (Drawbar Organ) topped the composite objective — a surprising centroid-dominated result on bass content. Cycle 2 ran a 180-cell fine fit around the top-5 coarse presets and returned an INDETERMINATE verdict leaning `RULED_OUT` for the SoundFont family: 90 of 180 renders (50%) collapsed to duplicate render SHAs, exposing an EQ-inertness defect in which post-processing choices had no audible effect. Cycle 3 rebuilt the fine-fit search into a v2 script that (a) drops zero-mean centring and LUFs-normalises every render before objective scoring so the EQ curve actually reaches the mel and VGGish features, (b) unconditionally promotes program 33 (Electric Bass Finger) alongside the stage-1 top-k so bass presets receive a fair hearing regardless of what stage-1 elevated, and (c) records the additional determinism dependencies (`pyloudnorm_available`, `lufs_target_db`) into a new `env_pin_sha256`. The v2 search was launched detached as a 216-cell sweep and completed at cycle-3 tail.

The stage-2b leaderboard resolves the two MODERATE-scoped questions carried out of cycle 2. The EQ v2 hypothesis is empirically confirmed: 215 of 216 rows now carry distinct render SHAs (up from 90 of 180 = 50% at cycle 2), so post-processing is no longer inert. Program

33 occupies the top seven slots of the composite ranking under fixed Electric-Bass-Finger tone (all EQ variants of the same preset). But under the embedding-cosine metric the top-5 is entirely program 19 (Church Organ) with $\text{embedding_cos_vggish} \approx 0.49$, well above program 33's best of 0.20. Under the pre-registered decision protocol this lands as `STILL_INDETERMINATE` — the composite and embedding metrics disagree on the winning preset. The formal family-verdict emission and the deterministic replay proof under the new env-pin fall to the next cycle by design; a launch cycle is bounded to setting up the falsification test, not running its conclusion.

1. Scope and framing

The v4 closure campaign covers six milestones in strict order: determinism certificate (M-V4-CERT), per-instrument sound profiles for five focus songs (M-V4-PROFILES), one full-song sound-matched showcase (M-V4-SHOWCASE), rules + lightweight-ear extractors (M-V4-RULES / M-V4-EAR), a seeded generator over five novel songs (M-V4-GEN), and a completion sweep (M-V4-CLOSE). Cycles 1–3 belong entirely to the first two milestones and, within M-V4-PROFILES, to the first song $\times$ instrument pair: Chicken Grease (`song_sha16 = 31a164f845f8e27e`) $\times$ bass.

Sound-matching is defined by the v4 spec as a two-phase policy: the per-instrument search may be stochastic and agentic, but the winning profile (parameter set + dependency hashes stored as `data/v4/profiles/<song_sha16>/<instrument>.json`) is pinned, and the replay of MIDI + profile $\rightarrow$ audio must be deterministic. Under operator relaxation dated 2026-09-03, byte-determinism is proved twice once per **render family** per song (SoundFont, stem-sampled, and bounce), not per profile — every profile still records its own render SHA for downstream re-verification when environment pins move.

The three fixed anchors preserved unchanged through cycles 1–3 are:

- General-MIDI SoundFont: `FluidR3_GM.sf2`, SHA 74594e8f4250680adf590507a306655a299935343583256f3b722c48a7
- Reference bass stem (`htdemucs_6s` separation of Chicken Grease): SHA 1bad871901294395c1b1ad1c97689e07
- Bass MIDI excerpt (from the canonical per-stem MuScriptor $\rightarrow$ canonical JSON $\rightarrow$ MIDI serializer): SHA 4863ca285c7db513c8bfc22da5e35e65036b0ecad2538a6d9794c80eb15f8ac9.

2. M-V4-CERT: the determinism certificate

The certificate at `docs/v3_determinism_certificate.md` was already complete on disk at the start of cycle 1. Its \$1 pipeline audit classifies all eleven stages of the v3 spine as either deterministic-proven (each running an internal byte-identity $\times 2$ gate on every invocation) or deterministic-by-construction (pure functions of pinned inputs with no PRNG). Zero at-risk stages remain.

Its \$2 records the end-to-end evidence: two runs of `scripts/v3_spine/recreate_v3_checkpointed.py --no-cache --verify-det` against Chicken Grease under the pinned session environment (`PYTHONHASHSEED=0`, `SOURCE_DATE_EPOCH=1756463424`, `TZ=UTC`, `LC_ALL=C.UTF-8`, single-thread OMP/MKL/OpenBLAS) produced byte-identical delivery WAVs across `original_ab`, `reconstruction_ab`, `full_reconstruction`, and every per-track file. The verdict recorded is `E2E_DETERMINISM_HOLDS` under `env_pin_sha256 = 623df01f262ffd180c8497ce9bb06a2d4438b9239d60dd997304830b6571d38d` (identical to the Peach Dream cycle-25 environment pin, confirming a stable session environment across songs). The certificate is re-issued only when the environment pin changes.

M-V4-CERT is therefore CONFIRMED. No cycle-1 work was required on it beyond the on-disk check that the campaign prompt asked for.

3. M-V4-PROFILES-1/cg-bass: cycle 1 — coarse SoundFont sweep

Cycle 1 ran the first-pass search over the General-MIDI SoundFont family via `scripts/sound_match/coarse_sweep_sf2.py`, launched detached through `_launch_cg_bass_sweep_c1.sh`. The grid was 15 GM presets against the reference bass stem, at the default gain / reverb / post configuration (gain 1.0, reverb send 0.3, no post-processing), scored by a fixed-weight composite:

$$\text{composite} = 0.5 \cdot \text{mel_L1_dB} + 0.25 \cdot \text{spectral_centroid_RMSE_Hz} + 0.25 \cdot (1 - \text{embedding_cos_vggish}) \cdot k$$

with objective weights frozen at the values shown in the profile `objective_scores.weights_frozen` block. Lower is better on all three components. Coarse-sweep leaderboard: program 17 (Drawbar Organ) topped the composite, ahead of the bass programs in the 32–39 range.

This is an unexpected result on bass content and was flagged as centroid-dominated: the organ’s smooth low-mid harmonic stack sits closer to the reference stem in mel-band L1 than the plucked electric-bass presets do, and the VGGish embedding component was not strong enough at the coarse grid to override it. The cycle-1 verdict was CONFIRMED for the coarse sweep as a stage-1 filter — its role is to hand a top-5 shortlist to fine-fit, not to name a winner — and the run advanced to cycle 2 for the fine fit.

4. M-V4-PROFILES-1/cg-bass: cycle 2 — fine fit v1 and the three MODERATE findings

Cycle 2 ran `scripts/sound_match/fine_fit_sf2.py` as an 180-cell sweep around the coarse top-5 presets, crossed with a 6-way gain grid (0.25–2.0), a 3-way reverb grid (0.0/0.3/0.7), and three post-processing modes (none, compressor_only, EQ_only, EQ_and_compressor). Two replay proofs were produced: the pinned c2 top-1 profile — program 17 organ at gain 0.5, reverb 0.3, no post-processing — reproduced byte-identically under a first `env_pin_sha256 = 2ac444c36298d6ada0579aba1a9160a5881703a4e628f5ccdd828b842a922ca`, and this was recorded as sub-milestone `cg-bass-sf2-replay-proof` at proof SHA 832868d0... with verdict `REPLAY_PROOF_HOLDS`. The profile UUID is 56cdc50a-dbbc-5a49-afc9-f3cf93a25c7d.

The cycle-2 auditor returned an overall verdict of INDETERMINATE leaning RULED_OUT for the SoundFont family on cg-bass and surfaced three MODERATE findings. Each names a specific defect and — because the auditor is adversarial — the exact evidence that would close it:

- **EQ inertness.** Of the 180 fine-fit renders, only 90 carried distinct `render_sha256` values: the post-processing chain was silently no-op, and the mel-band and VGGish objectives could not discriminate between EQ variants because the pipeline was normalising them away. The mechanism was zero-mean centring applied inside the scoring path after EQ but before mel/VGGish feature extraction — it cancelled the DC and low-frequency component that the EQ moved. Closure evidence would be: a re-run in which post-processing produces distinct SHAs for at least ~200 of ~216 cells.
- **Program 33 (Electric Bass Finger) absence.** The bass-family reference preset was not in the coarse top-5 and therefore never reached fine fit. The composite objective’s centroid-dominance was filtering it out at the coarse layer before the fine grid could give its post-processing a fair hearing. Closure evidence would be: program 33 unconditionally promoted into the stage-2b grid, and either ranking in the top-3 by embedding or providing empirical ground for elimination.
- **Profile-writer additive-extension audit.** The extended render-path parameters that

stage-2b intended to write into the profile schema (loudness_method, measured_db, applied_gain_lin, pyloudnorm_available, lufs_target_db) had to be added additively — that is, without changing the byte-for-byte identity of any already-pinned profile that omitted them. Closure evidence would be: a regression test proving that the cycle-2 top-1 profile’s SHA 11747a42cb1a8f7f... reproduces byte-identically when the writer runs under the extended signature.

Cycle 2 closed as CONFIRMED-with-findings: the fine fit itself ran cleanly and the replay proof held, but the composite objective was diagnosed as blind to the parameters it was ostensibly optimising over, and the search grid had failed to include the reference bass preset.

5. M-V4-PROFILES-1/cg-bass: cycle 3 — v2 rebuild, launch, and outcome

5.1 The v2 fine-fit script

Cycle 3 authored `scripts/sound_match/fine_fit_sf2_v2.py` as a sibling to the now-read-only `fine_fit_sf2.py`. Three changes address the cycle-2 findings by construction:

1. **Zero-mean centring is removed from the scoring path.** Each candidate render is instead loudness-normalised end-to-end using `pyloudnorm` at a fixed integrated-loudness target (`lufs_target_db`) before mel-band L1, spectral-centroid RMSE, and VGGish embedding cosine are computed. A recorded RMS-dBFS fallback path exists for environments without `pyloudnorm`; when the library is present it is used and its availability is stamped into the run’s environment pin.
2. **Program 33 (control_cell_electric_bass_finger) is unconditionally promoted** into the stage-2b grid alongside the stage-1 top-k, and the regression test `test_sound_match_fine_fit_sf2_v2.py` asserts that the grid always contains all 36 program-33 cells (9 gain × 3 reverb × 4 post variants — see §5.3) regardless of what stage-1 elevated.
3. **The environment pin absorbs the two new determinism dependencies.** The v1 pin `2ac444c36298d6ada0579abala9160a5881703a4e628f5cccd828b842a922ca` omitted `pyloudnorm_available` and `lufs_target_db`; the v2 pin incorporates both. This is an honesty choice under the campaign’s determinism doctrine: the loudness-normalisation library and its target are real dependencies of the render objective, and trimming them out of the pin to preserve equality with the v1 hash would pretend the search was reproducible when in fact swapping them would move every score.

`profile_writer.py` was extended in place — additively — to record the new fields; the cycle-2 top-1 profile SHA 11747a42cb1a8f7f... reproduces byte-identically under the extended signature (closure evidence for the third MODERATE finding, re-verified by the auditor in cycle 3 with a fresh 5/5 test run of `test_sound_match_profile_writer_v2.py`).

A diagnostic script `stage2b_zscore_diagnostic.py` was added to report per-cell z-scores across the leaderboard for downstream adjudication.

5.2 Launch and completion

The v2 sweep was launched detached via `_launch_cg_bass_stage2b_c3.sh`, recorded to `data/v4/logs/cg_bass_stage2b_c3.log` with PID 17998. At cycle-3 audit time 160 of 216 cell directories had materialised; at cycle-3 tail the full 216-cell leaderboard was written to `data/v4/profiles/31a164f845f8e27e/bass_stage2b/leaderboard.tsv`.

The cycle-3 auditor accepted the run under the launch-minimum contract because a cycle that launches detached and cleanly ends is a good cycle: bonus scope (the leaderboard and the family verdict) was permitted to defer into the next cycle. In fact the leaderboard did land in

cycle-3 tail, and this report reads from it directly below. The formal family-verdict emission and any fresh replay proof under the new environment pin still belong to the next cycle by design.

5.3 The stage-2b leaderboard

The completed sweep is 216 cells: 6 candidate presets (5 stage-1 top-k plus program 33 unconditionally promoted) $\times$ 9 gain steps $\times$ 3 reverb sends $\times$ 4 post-processing modes. Independent scan of the TSV:

- **Distinct render SHAs: 215 of 216** (99.5%). The EQ-inertness defect is closed empirically — post-processing is now audible in every cell but one (the single collision is a pair of cells whose compressor path was already inactive under the render’s dynamic range and reduces to none, mechanically explaining the tie).
- **Program 33 rows: 36 of 216** (unconditional-promotion invariant satisfied).

Top-5 by composite objective (lower is better):

rank	program	preset name	gain	reverb	post	mel L1 (dB)	emb. centroid cos		composite
							RMSE (Hz)	VG- Gish	
1	33	Electric Bass Finger	0.5	0.3	EQ_ only	7.87	1787	0.204	455.8
2	33	Electric Bass Finger	1.0	0.3	EQ_ only	7.87	1794	0.201	457.4
3	33	Electric Bass Finger	1.5	0.3	EQ_ only	7.87	1816	0.199	463.0
4	33	Electric Bass Finger	0.5	0.3	compressor_ only	7.46	1863	0.196	474.3
5	33	Electric Bass Finger	0.5	0.3	none	7.46	1863	0.196	474.3

Program 33 in fact holds the top seven composite slots and 10 of the top 13 — every configuration of the Electric-Bass-Finger preset in which reverb is 0.3 and gain is moderate outperforms every other preset on composite.

Top-5 by VGGish embedding cosine (higher is better):

rank	program	preset name	gain	reverb	post	emb. cos VGGish	composite
1	19	(Church Organ)	1.5	0.7	EQ_and_ compressor	0.495	712.4
2	19	(Church Organ)	0.5	0.0	EQ_and_ compressor	0.493	668.7

rank	program	preset name	gain	reverb	post	emb. cos VGGish	composite
3	19	(Church Organ)	1.0	0.7	EQ_and_ compressor	0.493	706.6
4	19	(Church Organ)	0.5	0.7	EQ_and_ compressor	0.491	693.7
5	19	(Church Organ)	1.0	0.0	EQ_and_ compressor	0.487	683.4

Program 33’s best embedding cosine (0.204) ranks 64th of 216.

5.4 Reading the verdict

The pre-registered adjudication protocol frames three outcomes for the SoundFont family on cg-bass:

- SF2_CONFIRMED iff top-1 embedding cosine ≥ 0.60 AND (program 33 in top-3 OR top-1 preset in {32..39}) AND spread $\geq 10\%$ relative to the cycle-1 coarse sweep.
- SF2 RULED_OUT iff top-1 embedding cosine < 0.40 AND program 33 not in top-5.
- STILL_INDETERMINATE otherwise.

The current leaderboard reads as STILL_INDETERMINATE. Top-1 embedding cosine is 0.495 — above the RULED_OUT threshold and below the CONFIRMED threshold. Program 33 sweeps the composite top-3 but does not appear in the embedding top-5 at all, and the top-1 by embedding is a non-bass organ preset. The two objective components disagree on the winner: the composite ranks by a mel-L1-dominated sum in which the plucked bass envelope wins, and the VGGish embedding ranks by mid-band spectral character in which the sustained organ wins.

This disagreement is itself the finding. The SoundFont family cannot be ruled out — Electric Bass Finger is now demonstrably the strongest plucked candidate under fair EQ and covers the full composite podium — but neither can it be declared confirmed at the operator’s listener-quality threshold on a VGGish score of 0.20. The disagreement was invisible in cycles 1-2: EQ inertness masked program-33’s post-processing sensitivity, and program-33’s absence from the stage-1 top-5 kept the plucked bass out of the fine fit entirely. Cycle-3’s v2 rebuild is what made both metrics legible at once.

6. Environment-pin scoping and the replay-proof carry-over

The env-pin change from cycle 2 to cycle 3 is not a defect but a correctness move under the campaign’s honesty rules. The consequence under the campaign’s replay-proof scoping is straightforward: the cycle-2 replay proof at 832868d0... continues to cover the cycle-2 pinned profile (56cdc50a-dbbc-..., program 17 organ) under its own environment pin 2ac444c36298d6ada..., unchanged. If the next cycle’s adjudication elects to pin a new cg-bass profile whose render was produced by the stage-2b sweep — for instance, one of the program-33 composite winners — that new profile requires its own replay proof under the new pin before it may carry the claim “SoundFont family LANDS for Chicken Grease bass.” The replay-proof module (replay_proof.py) completes in about 1.4 seconds of wall time; the cost is not the finding, the scope is.

If the next cycle elects to keep the cycle-2 profile pinned as-is (for instance, because the STILL_INDETERMINATE verdict argues for opening the stem-sampled family before over-committing on SoundFont), no re-emission is needed.

7. Open questions carried into cycle 4

Cycle 4 opens as an adjudication cycle, not a build cycle. Three items are queued:

1. **Emit the cg-bass family verdict.** Read the completed stage-2b leaderboard, run the pre-registered decision protocol, and write `data/v4/profiles/31a164f845f8e27e/bass_family_verdict.json`. On the current numbers this is `STILL_INDETERMINATE`; the protocol permits one more probe cycle before committing to the alternate render family by cost.
2. **Publish a full cg-bass sub-milestone report** covering cycles 1–3 in the sound-matching documentation directory.
3. **If a new cg-bass profile is pinned under the new environment pin, run a fresh SoundFont-family replay proof** and record the sub-milestone under `M-V4-PROFILES-1/cg-bass-sf2-replay-proof-v2`.

Beyond cycle 4, watch for the repeated-indeterminate signal: if a further probe cycle also returns `STILL_INDETERMINATE`, that is the campaign-level cue to open the stem-sampled builder over `data/v3/deliveries/31a164f845f8e27e/cert_run1/stems_6s/bass.wav` (SHA `1bad871901294395...`) as the alternate render family and let the cheaper family land the profile, rather than seeking further SoundFont refinements.

8. Two minor items noted but not acted on

- The `pyloudnorm` fallback path (RMS-dBFS scaling when the library is unavailable) is present in the v2 fine-fit script but was not exercised by the cycle-3 sweep, which ran with the library installed. Coverage for the fallback needs a future test cell if the environment ever regresses.
- The environment-pin expansion (adding `pyloudnorm_available` and `lufs_target_db` to the pin payload) is architecturally correct but has not yet propagated to the pipeline-wide env-pin manifest schema at `scripts/v3_spine/v3_pipeline/env_pin.py`. If v4 sound-matching adopts LUFs-integrated normalisation as a general render-path invariant, this schema alignment belongs in the campaign’s final cleanup milestone.

9. Conclusions

Cycles 1–3 land the determinism certificate trivially and take the first per-instrument sound-matching sub-milestone from a naive 15-preset coarse sweep through an inert-EQ diagnosis into a programmatically rebuilt search whose objective components are now legible and disagree in a specific, adjudicable way. The Chicken Grease bass profile is not yet finalised, but the search apparatus that will finalise it — a loudness-normalised v2 fine-fit with the reference bass preset unconditionally promoted, an additively extended profile writer whose byte-identity is regression-tested, and a stage-2b leaderboard whose 215 of 216 distinct render SHAs confirm post-processing is now audible — is complete. Cycle 4 adjudicates.

Appendix: Implementation details

A.1 Files created or extended (cycles 1–3)

- `scripts/sound_match/coarse_sweep_sf2.py` — cycle-1 coarse SoundFont sweep.
- `scripts/sound_match/fine_fit_sf2.py` — cycle-2 fine-fit v1 (now read-only).
- `scripts/sound_match/fine_fit_sf2_v2.py` — cycle-3 fine-fit v2 (LUFs normalisation, unconditional program-33 promotion).
- `scripts/sound_match/stage2_zscore_diagnostic.py` — cycle-2 diagnostic.
- `scripts/sound_match/stage2b_zscore_diagnostic.py` — cycle-3 diagnostic (v2).
- `scripts/sound_match/profile_writer.py` — extended additively cycle 3 with `loudness_method`,

measured_db, applied_gain_lin, pyloudnorm_available, lufs_target_db.

- scripts/sound_match/_launch_cg_bass_sweep_c1.sh, _launch_cg_bass_stage2_c2.sh, _launch_cg_bass_stage2b_c3.sh — detached launchers per cycle.
- tests/test_sound_match_fine_fit_sf2.py (cycle 2) — 8 tests.
- tests/test_sound_match_fine_fit_sf2_v2.py (cycle 3) — 14 tests.
- tests/test_sound_match_profile_writer_v2.py (cycle 3) — 5 tests, including byte-identity of cycle-2 profile SHA under extended signature.

Test totals at cycle-3 close: 19 of 19 passing across the two v2 suites (re-verified independently by the auditor).

A.2 Sweep artifacts

- Cycle-1 coarse leaderboard: data/v4/profiles/31a164f845f8e27e/bass_sweep_stage1/leaderboard.tsv, SHA 0623210a19de0c9602f0821827f5a6d1bba48097f3b99029500e22bf8f359b4f.
- Cycle-2 fine-fit leaderboard: data/v4/profiles/31a164f845f8e27e/bass_stage2/leaderboard.tsv, SHA 47aa8b0aca52ac85b1f1a1ff1b965f6602f30197774aa3acbfa2b11362bea278.
- Cycle-3 stage-2b leaderboard: data/v4/profiles/31a164f845f8e27e/bass_stage2b/leaderboard.tsv, 216 rows, 215 distinct render_sha256 values, 36 program-33 rows.
- Cycle-2 pinned profile: data/v4/profiles/31a164f845f8e27e/bass.json, profile UUID 56cdc50a-dbbc-5a49-afc9-f3cf93a25c7d, SHA 11747a42cb1a8f7f...
- Cycle-2 replay proof: data/v4/profiles/31a164f845f8e27e/bass.replay_proof.json, verdict REPLAY_PROOF_HOLDS, proof SHA 832868d0...

A.3 Environment pins in play across the arc

- env_pin_sha256 = 623df01f262ffd180c8497ce9bb06a2d4438b9239d60dd997304830b6571d38d — v3 spine driver session environment; the determinism certificate holds under this pin.
- env_pin_sha256 = 2ac444c36298d6ada0579abala9160a5881703a4e628f5ccdd828b842a922ca — cycle-2 fine-fit v1; cycle-2 cg-bass profile and replay proof anchor to this.
- env_pin_sha256 = <v2 hash, absorbing pyloudnorm_available + lufs_target_db> — cycle-3 fine-fit v2 environment; any new pinned profile from stage-2b anchors here and requires its own replay proof.

A.4 Fixed anchors preserved read-only across cycles 1-3

- scripts/palette_render/render_stem.py (SHA 214372d920a319a9...5b2b) — grep-verified untouched.
- Canonical JSON→MIDI serializer (cycle-4 anchor from the v3 arc) — untouched.
- Operator-blessed Chicken Grease Method A WAV (cycle-5 anchor from the v3 arc, SHA cc919559b4508b6bfe86...) — untouched.

A.5 Session references

Cycle 1: researcher 6da54f2f-faee-4c70-8de8-56ec8395705f, worker 27d5641d-a68c-4183-a222-f2fc7600da65, auditor 4f41efad-ea38-4f50-9989-42c6c33d9378. Cycle 2: researcher 543f0795-bf09-4116-a498-baf883014cac, worker b1e2dd48-9ce3-4d01-af02-d35474370bed, auditor be49cbaf-8e63-422a-877a-8d092138707c. Cycle 3: researcher 65c1fca9-5731-4ea6-bba7-bd47805d66f2, worker 86b63152-6182-458d-a216-821918be02b1, auditor f37ccaa6-aae4-47b4-a630-f4e0c142e3f6.

A.6 Cross-reference map

Cycle-1 coarse top-5 → cycle-2 fine-fit grid. Cycle-2 top-1 (program 17 organ, gain 0.5, reverb 0.3, none) → cycle-2 pinned profile 56cdc50a-... → cycle-2 replay proof 832868d0... Cycle-2 MODERATE findings (EQ inertness, program-33 absence, profile-writer additive extension) → cycle-3 architectural fixes in `fine_fit_sf2_v2.py` + additive fields in `profile_writer.py` → cycle-3 regression test byte-identity re-verification of cycle-2 profile SHA under extended signature. Cycle-3 v2 sweep → 216-row stage-2b leaderboard → cycle-4 family-verdict emission (open).